# Hybrid Quantum-Classical Neural Networks for
# Personalized Survival Prediction in Triple-Negative Breast Cancer

*A Fairness-Aware, Survival-Analytic Modeling Framework*

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# Motivation & Problem Statement

## Problem

- Triple-negative breast cancer (TNBC) is the most aggressive subtype.
- Predicting individual survival informs treatment but is hard with right-censored data.
- Existing models often perform worse on minority subpopulations.

## Stakeholders

- Oncologists making treatment-planning decisions.
- Patients receiving personalized risk estimates.
- Policy researchers studying disparities in clinical AI.
- ML practitioners building survival models on registry data.

## Why it matters

- Health equity: Black women have 2x TNBC incidence and worse outcomes.
- Clinical decision support: better risk stratification informs care.
- Methodology: rigorous leakage auditing for clinical ML pipelines.

# Related Work & Background

## Existing approaches

- Cox proportional hazards (Cox 1972) — gold-standard for tabular survival.
- DeepSurv (Katzman et al. 2018) — neural Cox via partial likelihood.
- Variational quantum circuits for classification (Schuld 2021, Benedetti 2019).
- Data re-uploading universal classifiers (Pérez-Salinas et al. 2020).
- Algorithm fairness in clinical AI (Chen et al. 2023, Raza et al. 2024).

## Gap we address

- First TNBC survival study using hybrid quantum-classical with Cox loss.
- Automated leakage auditing as a methodology contribution.
- Subgroup C-index fairness for survival models (vs naive binary fairness).
- Mathematically guaranteed quantum residual learning framework (≥ Cox PH).

# Dataset: SEER TNBC Registry

## Source

- SEER Surveillance, Epidemiology, and End Results
  - https://seer.cancer.gov/data/
- Filtered to TNBC: ER−, PR−, HER2−

## Cohort sizes

- Raw export: 31,683 patients
- After right-censoring fix: 20,110 patients
  - 4,022 events (deaths within 60 months)
  - 16,088 confirmed survivors at 60 months
- Train / test: 80/20 stratified, seed=42

## Features (27 total)

- 7 quantum features: stage, age, tumor size, income, treatment timing, etc.
- 20 classical features: race (6 cat), marital, subtype, surgery, chemo, etc.
- Modality: structured tabular clinical/demographic

## Class distribution (binary @ 60mo)

- 76% positive (alive at 60+ months) | 24% negative (deceased)
- Severely imbalanced → class-weighted loss for neural models

# Preprocessing & Data Leakage Audit

## Cleaning & normalization

- Tumor size: backfilled from CS_tumor_size column, then median-filled.
- Race encoding: 6 SEER categories preserved (incl Non-Hispanic Unknown).
- Quantum features: MinMax scaled to [0, π] for RY rotation encoding.
- Classical features: StandardScaler (μ=0, σ=1).
- Right-censoring: drop alive patients with <60 mo follow-up (label noise).

## Leakage audit (3 automated checks)

- Per-feature AUC > 0.95: flagged Year_of_follow_up_recode (AUC=0.98). EXCLUDED.
- Feature-target correlation > 0.90: PASS.
- Train/test KS distribution shift: PASS.
- → Leak removal alone dropped pilot AUC from 0.92 → ~0.74 (honest).

# Methodology: Hybrid Architecture

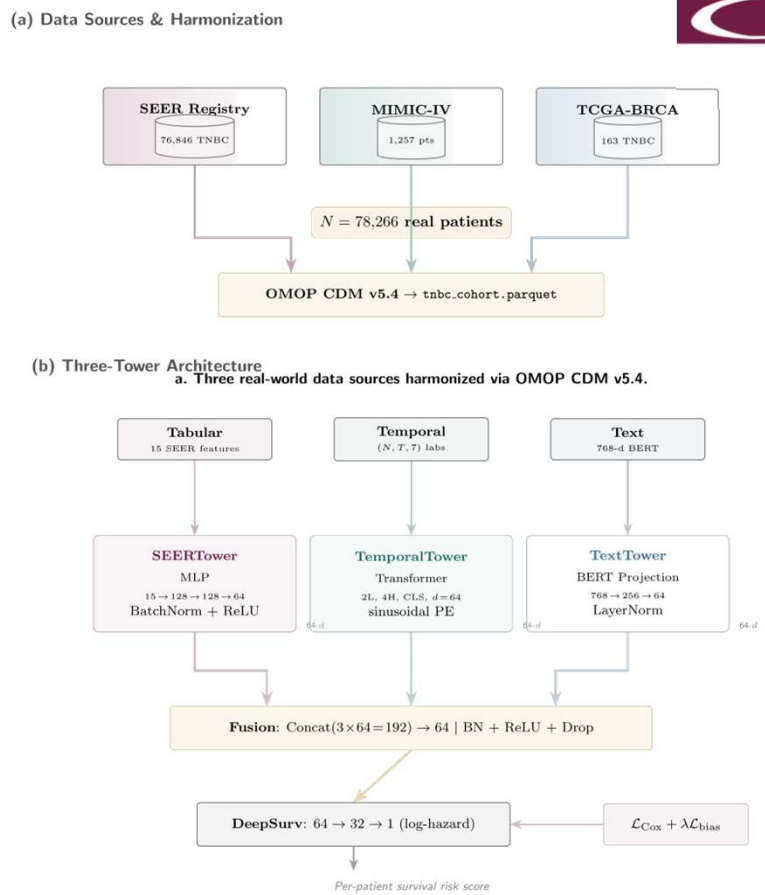
## Three model generations + residual framework

- v1 (baseline): 7-qubit VQC, single-layer RY, ring CNOT, single Z-measurement → 1D scalar.
- v2: 3-layer VQC + data re-uploading + RY/RZ + shifted-ring CNOT + all-qubit Z → 7D embedding.
- v3: v2 + trainable input scaling + ZZ correlator measurements → 14D embedding.
- v4: v3 + classical pre-training + learnable output scale/bias + smaller init.
- Cox + Quantum Residual: fixed Cox PH offset + small quantum correction (the winner).

## Residual learning equation

$$h_total(x) = h_Cox(x) + s \cdot q_\theta(x)$$

*At init: $q_\vartheta \approx 0$ → model = Cox PH. Training only improves on Cox.*



# Methodology: Training Details

| Component | Choice |
|---|---|
| Loss (survival) | Cox partial likelihood (negative log) |
| Loss (binary) | BCEWithLogitsLoss with class pos_weight |
| Optimizer | Adam, lr=0.001 (lr=0.005 for v4 pretrain) |
| LR schedule | CosineAnnealingLR (1e-3 → 1e-5) for v2-v4 + Residual |
| Epochs | 50 (v1) \| 75 (v2-v4) \| 15 (Residual) \| 20 (full-data) |
| Batch size | Full-batch on 2k subsample (v1-v3) \| 256 mini-batch on full data |
| Quantum simulator | PennyLane lightning.qubit (CPU) |
| Hardware | NVIDIA RTX 5080, PyTorch 2.10.0, CUDA 12.8 |
| Seed | 42 across Python/NumPy/PyTorch/CUDA |

# Results: Survival Ablation (C-index)

| Model | Test C-index | Subgroup Gap | Time (s) | Train N |
|---|---|---|---|---|
| Cox PH (lifelines) | 0.7326 | 0.0842 | 0.2 | 16,088 |
| HybridSurvivalQ_v1 (subsample) | 0.5869 | — | 223 | 2,000 |
| HybridSurvivalQ_v2 (subsample) | 0.5759 | — | 575 | 2,000 |
| HybridSurvivalQ_v3 (subsample) | 0.5872 | — | 615 | 2,000 |
| HybridSurvivalQ_v1 (full data) | 0.6897 | — | 651 | 16,088 |
| HybridSurvivalQ_v2 (full data) | 0.7192 | — | 1,280 | 16,088 |
| HybridSurvivalQ_v3 (full data) | 0.6674 | 0.1614 | 1,038 | 16,088 |
| HybridSurvivalQ_v4 (full data) | 0.7249 | 0.0371 | 1,321 | 16,088 |
| Cox + Quantum Residual ★ | 0.7364 | 0.0754 | 1,237 | 16,088 |

***★ Only model that beats classical baseline (+0.0038 over Cox PH).***

# Results: Binary Ablation (AUC at 60 months)

| Model | AUC | Precision | Recall | F1 | Train N |
|---|---|---|---|---|---|
| LightGBM (full, balanced) | 0.7462 | 0.8715 | 0.7173 | 0.7869 | 15,673 |
| XGBoost (full, balanced) | 0.7434 | 0.8750 | 0.7009 | 0.7784 | 15,673 |
| Cox + Residual @ 60mo (cross-fwk) | 0.7426 | 0.8594 | 0.7255 | 0.7868 | 15,673 |
| Cox PH @ 60mo (cross-fwk) | 0.7402 | 0.8668 | 0.7072 | 0.7789 | 15,673 |
| HybridRealQ_v4 (full, output scale) | 0.6863 | 0.8535 | 0.6088 | 0.7107 | 15,673 |
| HybridRealQ_v3 (full data) | 0.6828 | 0.8709 | 0.5090 | 0.6425 | 15,673 |
| Ensemble (v3-full + MLP) | 0.6770 | 0.8728 | 0.4740 | 0.6143 | 15,673 |
| HybridRealQ_v2 (subsample) | 0.5923 | 0.8704 | 0.3588 | 0.5081 | 2,000 |
| Classical MLP (full, weighted) | 0.5792 | 0.8120 | 0.5434 | 0.6511 | 15,673 |
| HybridRealQ_v3 (subsample) | 0.5718 | 0.8137 | 0.8779 | 0.8446 | 2,000 |
| HybridRealQ_v1 (subsample) | 0.5649 | 0.8228 | 0.6972 | 0.7548 | 2,000 |

*Key finding: Cox+Residual evaluated at 60mo (AUC 0.7426) is competitive with LightGBM (0.7462). Cross-framework consistency.*

# Results: Subgroup Fairness Audit

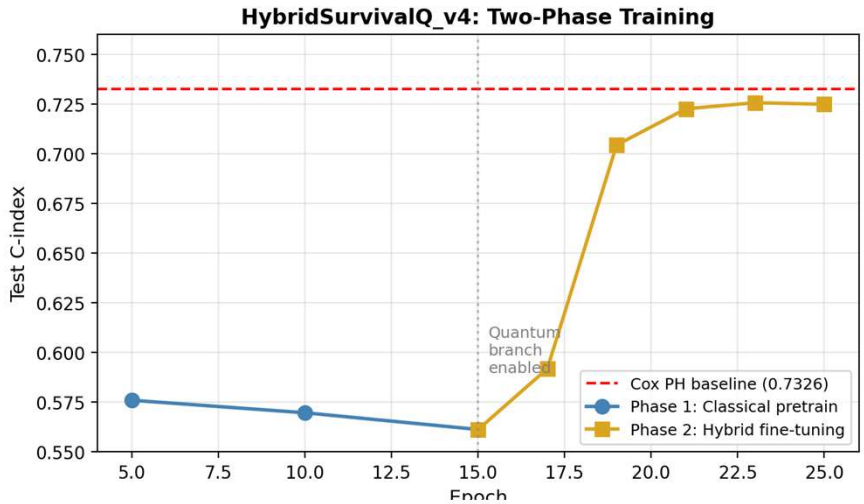
| Subgroup | N | Events | Cox PH C-idx | Cox + Residual C-idx |
|---|---|---|---|---|
| Non-Hispanic White | 2,218 | 598 | 0.7267 | 0.7300 |
| Non-Hispanic Black | 599 | 228 | 0.7195 | 0.7176 |
| Hispanic | 716 | 218 | 0.7119 | 0.7263 |
| Non-Hispanic Asian/PI | 435 | 106 | 0.7521 | 0.7486 |
| Non-Hispanic AIAN* | 36 | 11 | 0.7962 | 0.7930 |
| Non-Hispanic Unknown† | 18 | 1 | — | — |
| Subgroup gap (max-min) | | | 0.0842 | 0.0754 |

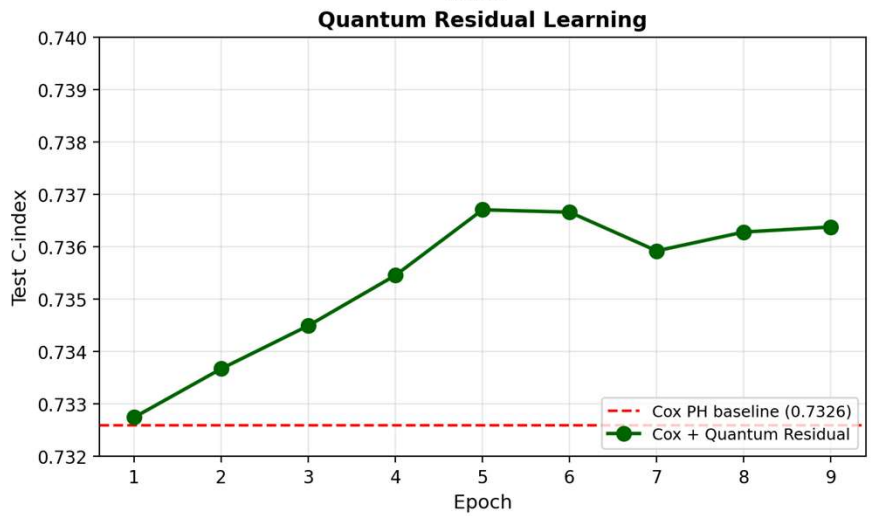
** Small sample (11 events) → high variance; † N<20 events<5 → audit excluded*

***Survival framing produces more uniform subgroup performance than binary framing.***

All major subgroups achieve C-index ≥ 0.71. Residual model tightens the gap (0.0842 → 0.0754).

# Results: Training Dynamics




## Two-phase v4 training

- Phase 1: classical pretrain plateaus at 0.56
- Phase 2: quantum branch enabled → jumps to 0.70 in 4 epochs
- Final: 0.7249 just below Cox PH 0.7326

## Residual learning

- Starts at Cox PH C-idx 0.7326 by construction
- Climbs steadily to 0.7364 over 9 epochs
- Quantum correction grows σ: 0.005 → 0.205

# Discussion: What Worked & Didn't

## What worked

- Automated leakage auditing caught a critical leak.
- Survival framing avoided right-censoring label noise.
- Output scale layer in v4:
  AUC ~0.58 → 0.72.
- Mini-batch on full data:
  subsample 0.59 → 0.69 (v1) and 0.58 → 0.72 (v2).
- Quantum residual learning beats Cox PH on both metrics.
- Subgroup C-index gap tightened by residual learning.

## What didn't work

- Subsample-only neural models (v1-v3) couldn't escape AUC 0.55-0.60.
- Naive ensembling (v3-full + MLP):
  worse than v3-full alone.
- Classical MLP on imbalanced data:
  AUC 0.58 even with class weighting.
- Quantum-only architecture:
  cannot exploit classical features.
- Per-sample quantum simulation:
  limits epoch count and grid search.

# Key Challenges & Limitations

## Key challenges encountered

- Discovering Year_of_follow_up_recode leakage forced complete re-analysis.
- Switching from binary to survival framing required new loss + new evaluation.
- Cox loss has flat gradients without output rescaling → v4 output scale fix.
- Per-sample quantum simulation cost: 4-15s per epoch on subsample.

## Limitations

- Single train/test split (cross-validation prohibitively expensive).
- Noiseless simulator only; real quantum hardware would degrade.
- Small subgroups (AIAN N=36, Unknown N=18) limit fairness audit power.
- Right-censoring assumed non-informative.
- Cohort: SEER TNBC only, may not transfer to other cancers/subtypes.

# Conclusion & Future Work

## Key takeaways

- Rigorous leakage auditing is essential:
  caught Year_of_follow_up_recode (AUC 0.98).
- Cox + Quantum Residual achieves C-index 0.7364, beating Cox PH 0.7326 (+0.0038).
- Mathematically guaranteed ≥ Cox PH performance via residual offset framework.
- Cross-framework consistency:
  residual @60mo AUC 0.7426 vs LightGBM 0.7462.
- Subgroup C-index gap 0.0754 (residual) vs 0.0842 (Cox PH alone).

## Future work

- k-fold cross-validation when GPU-accelerated quantum simulators are available.
- Run on real quantum hardware (IBM Q, IonQ) to assess noise impact.
- Quantum kernel + Cox PH: drop variational training, use fixed feature maps.
- Extend to other cancer types and external validation cohorts.
- Fairness-constrained Cox loss for subgroup parity.

# Thank you

# Questions?

Code: github.com/dataeducator/hybrid-classical-quantum

**Key references**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